

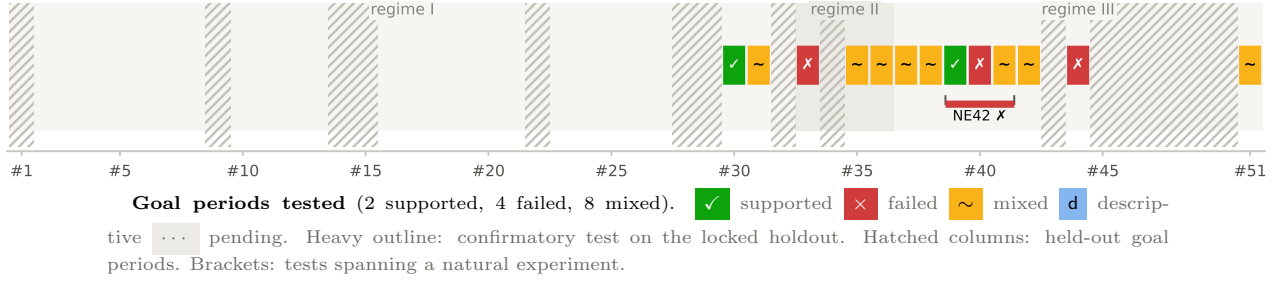
H128: Free kickoffs coarsen, named kickoffs freeze: a kinetic Potts quench

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1/5 Usefulness 1.5/5

Bottom line. After a goal kickoff, the spread of agents over work projects does not separate free from named kickoffs (freeze time $p = 0.32$, excess area $p = 0.73$, power ≥ 0.96), and free weeks lose projects by finishing, not merging (merge share 0.00–0.38): HH369's coarsening-vs-freeze split is killed.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

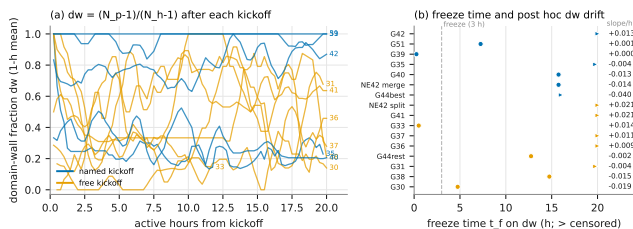
HH369 says that a goal kickoff is a zero-temperature Potts quench on the agents' work projects. A free kickoff should coarsen: projects merge and their number falls as $t^{-\alpha}$ with $\alpha \approx 0.3$ –0.5. A kickoff that names its target should freeze at once. H128 replays every agent's work-ledger host label after 12 non-holdout kickoffs (7 free, 5 named; regimes I–III) and two natives (#44 rooms, NE42). **Free and named kickoffs give the same curves:** freeze time $p = 0.32$, excess area $p = 0.73$, where a true coarsening-vs-freeze world gives $p < 0.05$ with power 0.96–1.00. Free weeks lose projects by finishing, not merging (merge share 0.00–0.38 in 7/7). HH369's kill is met. The holdout is not run.

PHYSICAL MODEL

Model 10, kinetic Potts quench with births. At each of its work windows an agent picks repo j with $P(j) \propto e^{\beta J s_j + h_j + b[j=\sigma_i]}$ plus a new-repo option. Free kickoff: $h = 0$, so domains merge by coupling (mean-field voter: $N_p \propto 1/t$). Named kickoff: $h_T \gg 1$ on a shared target or one target per agent, so the count jumps to its final value in one round. Rival R1: projects end by finishing (expiry), with no merging.

$$d_w(t) = \frac{N_p(t) - 1}{N_h(t) - 1}, \quad t_f = \min\{t : |d_w(s) - d_w^\infty| \leq \delta \forall s \geq t\}, \quad \delta = \max\left(\frac{1}{N_h - 1}, 0.15 d_w^\infty\right)$$

$$A_u = \int_{t_{pk}}^{t_{pk} + 8h} \frac{d_w(t) - d_w^\infty}{d_w^{pk} - d_w^\infty} dt, \quad c_m = \frac{\# \text{ merge deaths}}{\# \text{ project deaths}}$$



Observables. (a) Domain-wall fraction d_w after each kickoff (1-h running mean; named blue, free orange). Per-agent named weeks (#39, #51) sit at $d_w \approx 1$; free weeks stay scattered, fragment (#41) or drift slowly. (b) Freeze time t_f per unit (triangles: censored at the horizon), with the post hoc drift of d_w per active hour. The dashed line is the 3-h freeze criterion. The kickoff code does not order either quantity.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator: naming the target in a kickoff does not make the swarm converge on fewer projects. Own-artifact goals freeze at one project per agent from the first hour. A free goal leaves agents scattered or splitting, and projects end when work stops, not by merging (merge share ≤ 0.38). To get agents onto one repo, give a shared target to a swarm that is already scattered (NE42: d_w 1 to 0.2 over about 16 active hours). Expect hours, not minutes.

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
F: rule valid on skeletons	d_w 12/13; HH's N_p 0/16 (A1)	d_w only
P1 free: $t^{-\alpha}$, $\alpha \in 1/7$ (#30, $\hat{\alpha}$ 0.44 [0.32, 0.60])		failed
P2 named: freeze ≤ 3 h	1/5 (#39, frozen at $d_w = 1$)	failed
P3 free slower (kill test)	t_f $p = 0.32$; A_u $p = 0.73$	killed
P4 merging, not finishing	c_m 0.00–0.38 in 7/7 free	failed
P5 power beats exponential	3/7	failed
N1 #44 rooms	free room flat at 0.92; named drifts	failed
N2 NE42 A-B-A	named merge coarsens (0.26); failed free splits	
N3 #37 replicas	one room < 2 hosts	descr.

GOAL PERIODS IN DETAIL

G30 ✓ supported

G31 ~ mixed

G33 × failed

G35 ~ mixed

G36 ~ mixed

G37 ~ mixed

G38 ~ mixed

G39 ✓ supported

G40 × failed

G41 ~ mixed

+4 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The observable was changed before real data (A1): N_p failed the synthetic rule on every skeleton. Power law vs exponential is not identified at village sizes (a true coarsening selects the power law in $\leq 48\%$ of runs), so P1 and P5 were weak by design; the kill contrast P3 is the powered test. Host labels lapse after 100 own calls, so “finishing” mixes shipped and abandoned projects. The NE42 coarsening reading and the drift signs are post hoc (A2). #44 #best has 4 hosts (below the testability floor).

ROUND 2

Measure field strength per agent (share of first post-kickoff commits on the settled repo), split finishing into shipped vs dropped with deploy commits, and test “shared target into a scattered start” on held-out #45. Confirm C1–C3 on #46–#49.

- **H128-R1** A per-agent field strength, not a kickoff code.
- **H128-R2** Separate finishing from abandonment.
- **H128-R3** Shared target into a scattered start.

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Second figure. Synthetic validation on the real skeletons (60 runs per world and period). (a) The primary d_w classifies coarsening worlds as slow and field or finishing worlds as frozen. (b) The HH’s literal project count N_p calls per-agent freezes and pure finishing “slow” in every run: it tracks how many agents hold a label, not the number of domains (Amendment A1).

